Lesson plan · for the teacher

Lists: a tool for your marks

Programming · Level 2 · Lesson 4

Time
40 min

Materials
Board and chalk; Ten made-up marks (never real pupils' marks); Your phone with this lesson downloaded

Learning goals

- Keep marks in a list and find them by position.
- Use len, sum, max and min.
- Loop over a list to count and average.

Activity: A human list

1. Warm-up (5 min): Five pupils stand in a row, each holding a made-up mark. Write marks = [...] on the board in the same order. Ask: how could we talk about one mark without a name?
2. Teach (10 min): Label the pupils 0 to 4 and explain that Python counts from 0. Write len, sum, max and min, and work out each one with the row.
3. Ask the list (10 min): Call out marks[2], marks[-1], max(marks). The pupil who holds the answer steps forward; the class checks.
4. Compare (10 min): Groups work out the average by hand while you run sum(marks) / len(marks) on your phone. Compare.
5. Wrap-up (5 min): Ask why counting from 0 causes bugs, and what marks[5] would do. Then give the exit ticket.

Use made-up marks only. Never put real pupils' names or marks on a shared board or a shared phone.

Discussion

- Why do you think lists count from 0?
- What would sorted(marks) tell you that max(marks) doesn't?
- Which list in our school would you like Python to summarise?

Exit ticket

- Write a list of three marks and the line that prints their average.

Try it at school tomorrow
Write ten made-up marks as a Python list on the board. Groups find the average and the top mark by hand while you run sum(marks) / len(marks) and max(marks) on your phone. Who finished first, and did you agree?

Practice questions · for you

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1 **What does this program print?**

```
marks = [10, 12, 14]
print(sum(marks) / len(marks))
```

◯ 12.0 ◯ 3 ◯ 36

2 **The file marks.csv has a heading line, then one pupil per line, such as Amina,14. Finish the program: it asks for the pass mark and prints how many pupils reached it, like Passed: 3.**

```
pass_mark = int(input("Pass mark? "))
passed = 0
with open("marks.csv") as f:
    f.readline()  # skip the heading
    for line in f:
        name, mark = line.strip().split(",")
        # count the pupils who reached pass_mark
```

Write your program:

3 **Put the lines in order: the program prints every mark below 10, then the class average. One line doesn't belong.**

Number the lines in order (1, 2, 3...), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `print("Average:", sum(marks) / len(marks))`

☐ `marks = [14, 9, 17, 6]`

☐ `for m in range(marks):`

☐ `for m in marks:`

☐ `print("Needs help:", m)`

☐ `if m < 10:`

4 **This program should print the last of the three marks, but it stops with an IndexError. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  marks = [14, 9, 17]
2  last = marks[3]
3  print("Last mark:", last)
```

5 **marks = [12, 15, 9]. Which of these give the class average, 12.0?**

☐ (12 + 15 + 9) / 3 ☐ max(marks) - min(marks) ☐ sum(marks) / len(marks) ☐ sum(marks) / 2

✎ **Which list do you keep by hand (marks, attendance, book loans)? What would you ask Python to tell you about it: the average, the top mark, who needs help?**

Check your answers

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1 **12.0**

sum(marks) is 36 and len(marks) is 3; 36 / 3 is 12, and / always gives a decimal, so Python prints 12.0.

2

```
pass_mark = int(input("Pass mark? "))
passed = 0
with open("marks.csv") as f:
    f.readline()  # skip the heading
    for line in f:
        name, mark = line.strip().split(",")
        if int(mark) >= pass_mark:
            passed = passed + 1
print("Passed:", passed)
```

Each line of the file is split at the comma into a name and a mark.

3 **Top to bottom: 5, 1, ✗, 2, 4, 3**

```
marks = [14, 9, 17, 6]
for m in marks:
    if m < 10:
        print("Needs help:", m)
print("Average:", sum(marks) / len(marks))
```

for m in marks: visits each mark itself.

4 **Line 2: `last = marks[2]`**

Three marks sit at positions 0, 1 and 2, so there is no marks[3].

5 **sum(marks) / len(marks); (12 + 15 + 9) / 3**

The average is the total divided by how many marks there are: 36 / 3.